

Haimin Hu

CONTACT INFORMATION	3400 N Charles St Baltimore, MD 21218, United States U.S. Permanent Resident	Alliance Lab  Google Scholar  haimin@cs.jhu.edu 
RESEARCH INTERESTS	My vision is to enable human-centered robotic systems that can be scaled, verified, and trusted. Towards this goal, I work on new algorithms, theorems, and systems, drawing on principles from dynamic game theory, reinforcement learning, foundation models, and robot safety. I envision a future where humans and robots learn and grow together.	
PROFESSIONAL POSITIONS	Johns Hopkins University , Baltimore, MD Assistant Professor, Department of Computer Science Director, Alliance Lab Affiliations: Data Science and AI Institute , Laboratory for Computational Sensing and Robotics , Institute for Assured Autonomy	2026 - present
PAST POSITIONS	University of Pennsylvania , Philadelphia, PA Postdoctoral Fellow, GRASP Lab , PRECISE Center Toyota Research Institute , Cambridge, MA PhD Research Intern, Human-Interactive Driving Team Honda Research Institute USA, Inc. , San Jose, CA PhD Research Intern National Institute for Nuclear Physics & University of Padova , Padova, Italy Machine Learning Research Intern	2025 - 2026 2023 2022 2017
EDUCATION	Princeton University , Princeton, NJ <i>Ph.D. in Electrical and Computer Engineering</i> <i>M.A. in Electrical and Computer Engineering</i> Advisor: Jaime Fernández Fisac Main collaborators: Naomi Ehrich Leonard, Bartolomeo Stellato, Andrea Bajcsy Thesis: Game-Theoretic Integration of Safety and Learning for Human-Centered Robotics University of Pennsylvania , Philadelphia, PA <i>M.S.E. in Electrical Engineering</i> Advisors: George Pappas, Manfred Morari, Nikolai Matni University of California, Berkeley , Berkeley, CA <i>Visiting student in Electrical Engineering and Computer Sciences</i> Advisor: Claire Tomlin ShanghaiTech University , Shanghai, China <i>B.E. in Electronic and Information Engineering</i> Advisor: Boris Houska	2020 - 2025 2020 - 2022 2018 - 2020 2017 - 2018 2014 - 2018
HONORS AND AWARDS	Penn AI Fellow Robotics: Science and Systems Pioneer <i>"30 world's top early-career researchers in robotics."</i> Cyber-Physical Systems Rising Star <i>"30 outstanding PhD students and postdocs in Cyber-Physical Systems (CPS)."</i> Human-Robot Interaction Pioneer <i>"25 world's top student researchers in human-robot interaction (HRI)."</i>	University of Pennsylvania, 2026 RSS Foundation, 2025 NSF, 2025 ACM/IEEE, 2024

Roberto Tempo Best CDC Paper Nominee	IEEE Control Systems Society, 2024
Best Presentation Award	RSS Safe Autonomy Workshop, 2024
Best Poster Award	Workshop on Optimization, Learning, and Control, 2024
Outstanding Presentation Award	Princeton Research Day, 2024
<i>Awarded to 1 group of graduate students across all majors.</i>	
Bede Liu Fund for Excellence (Travel Grant)	Princeton University, 2022 - 2025
Outstanding Research Award	University of Pennsylvania, 2020
<i>Awarded to 2 electrical engineering masters.</i>	
Outstanding Graduate	Shanghai Municipal Education Commission, 2018
President's Scholarship	ShanghaiTech University, 2017
Merit Student for Excellence in Research	ShanghaiTech University, 2017
<i>Awarded to 1 undergraduate across all majors.</i>	
Scholarship for Overseas Study	ShanghaiTech University, 2016 - 2018
Scholarship for Academic Excellence	ShanghaiTech University, 2015
Member of Sigma Xi	

JOURNAL PUBLICATIONS

- [J4] **H. Hu**, D. Isele, S. Bae, and J. F. Fisac. [Active Uncertainty Reduction for Safe and Efficient Interaction Planning: A Shielding-Aware Dual Control Approach](#), *The International Journal of Robotics Research (IJRR)*, vol. 43, no. 9, pp. 1382–1408, 2024 (first published online 2023).
- [J3] K.-C. Hsu, **H. Hu**, and J. F. Fisac. [The Safety Filter: A Unified View of Safety-Critical Control in Autonomous Systems](#), *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 7, pp. 47–72, 2024.
- [J2] **H. Hu**, K. Nakamura, and J. F. Fisac. [SHARP: Shielding-Aware Robust Planning for Safe and Efficient Human-Robot Interaction](#), *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 2, pp. 5591–5598, 2022.
- [J1] M. Chen*, S. L. Herbert*, **H. Hu**, Y. Pu, J. F. Fisac, S. Bansal, S. Han, and C. J. Tomlin. [FaSTrack: A Modular Framework for Real-Time Motion Planning and Guaranteed Safe Tracking](#), *IEEE Transactions on Automatic Control (TAC)*, vol. 66, no. 12, pp. 5861–5876, 2021.

CONFERENCE PUBLICATIONS

- [C23] Y. Sun et al. (including **H. Hu**). [Agents' Last Exam](#), *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- [C22] W. Wang, E. Gu, A. Loquercio, **H. Hu**[†], and R. Mangharam[†]. [AI Coaching for Accelerating Human Skill Development with Reinforcement Learning](#), *Conference on Robot Learning (CoRL)*, 2026.
- [C21] D. D. Oh, D. P. Nguyen, **H. Hu**, and J. F. Fisac. [Synthesis and Deployment of Maximal Robust Control Barrier Functions through Adversarial Reinforcement Learning](#), *IEEE Conference on Decision and Control (CDC)*, 2026.
- [C20] **H. Hu**. [Permissive Safety Through Trusted Inference: Verifiable Belief-Space Neural Safety Filters for Assured Interactive Robotics](#), *World Symposium on the Algorithmic Foundations of Robotics (WAFR) XVII*, 2026.
- [C19] K. Liang, **H. Hu**, R. Liu, T. L. Griffiths, and J. F. Fisac. [RLHS: Mitigating Misalignment in RLHF with Hindsight Simulation](#), *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 11457–11483.
- [C18] D. D. Oh*, D. P. Nguyen*, **H. Hu**, and J. F. Fisac. [Provably Optimal Reinforcement Learning under Safety Filtering](#), *Proceedings of the IASEAI Conference*, vol. 2, no. 1, pp. 488–500, 2026.

* indicates equal contribution; † indicates equal advising.

- [C17] D. D. Oh*, J. Lidard*, **H. Hu**, H. Sinhar, E. Lazarski, D. Gopinath, E. Sumner, J. DeCastro, G. Rosman, N. E. Leonard, and J. F. Fisac. [Safety with Agency: Human-Centered Safety Filter with Application to AI-Assisted Motorsports](#), *Robotics: Science and Systems (RSS)*, 2025.
- [C16] **H. Hu**, J. F. Fisac, N. E. Leonard, D. Gopinath, J. DeCastro, and G. Rosman. [Think Deep and Fast: Learning Neural Nonlinear Opinion Dynamics from Inverse Dynamic Games for Split-Second Interactions](#), *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [C15] J. Wang*, **H. Hu***, D. P. Nguyen, and J. F. Fisac. [MAGICS: Adversarial RL with Minimax Actors Guided by Implicit Critic Stackelberg for Convergent Neural Synthesis of Robot Safety](#), *Workshop on the Algorithmic Foundations of Robotics (WAFR) XVI*, 2024.
- [C14] **H. Hu***, G. Dragotto*, Z. Zhang, K. Liang, B. Stellato, and J. F. Fisac. [Who Plays First? Optimizing the Order of Play in Stackelberg Games with Many Robots](#), *Robotics: Science and Systems (RSS)*, 2024.
- [C13] J. Lidard*, **H. Hu***, A. Hancock, Z. Zhang, A. G. Contreras, V. Modi, J. DeCastro, D. Gopinath, G. Rosman, N. E. Leonard, M. Santos, and J. F. Fisac. [Blending Data-Driven Priors in Dynamic Games](#), *Robotics: Science and Systems (RSS)*, 2024.
- [C12] **H. Hu**. [Doxo-Physical Planning: A New Paradigm for Safe and Efficient Human-Robot Interaction under Uncertainty](#), *Companion Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI Pioneers)*, pp. 106–108, 2024.
- [C11] **H. Hu***, Z. Zhang*, K. Nakamura, A. Bajcsy, and J. F. Fisac. [Deception Game: Closing the Safety-Learning Loop in Interactive Robot Autonomy](#), *Conference on Robot Learning (CoRL)*, 2023. **Best Presentation Award** at the 2024 RSS Safe Autonomy Workshop.
- [C10] **H. Hu**, K. Nakamura, K.-C. Hsu, N. E. Leonard, and J. F. Fisac. [Emergent Coordination Through Game-Induced Nonlinear Opinion Dynamics](#), *IEEE Conference on Decision and Control (CDC)*, 2023. **Nominated for the Roberto Tempo Best CDC Paper Award**.
- [C9] **H. Hu** and J. F. Fisac. [Active Uncertainty Reduction for Human-Robot Interaction: An Implicit Dual Control Approach](#), *World Symposium on the Algorithmic Foundations of Robotics (WAFR) XV*, 2022. **Invited extension for IJRR special issue**.
- [C8] L. Lindemann, **H. Hu**, A. Robey, H. Zhang, D. V. Dimarogonas, S. Tu, and N. Matni. [Learning Hybrid Control Barrier Functions from Data](#), *Conference on Robot Learning (CoRL)*, 2020 (proceedings published 2021).
- [C7] **H. Hu**, M. Fazlyab, M. Morari, and G. J. Pappas. [Reach-SDP: Reachability Analysis of Closed-Loop Systems with Neural Network Controllers via Semidefinite Programming](#), *IEEE Conference on Decision and Control (CDC)*, 2020.
- [C6] A. Robey*, **H. Hu***, L. Lindemann, H. Zhang, D. V. Dimarogonas, S. Tu, and N. Matni. [Learning Control Barrier Functions from Expert Demonstrations](#), *IEEE Conference on Decision and Control (CDC)*, 2020.
- [C5] **H. Hu**, K. Gatsis, M. Morari, and G. J. Pappas. [Non-Cooperative Distributed MPC with Iterative Learning](#), *21st IFAC World Congress*, 2020.
- [C4] **H. Hu**, K. Gatsis, M. Morari, and G. J. Pappas. [Tuning Communication Latency for Distributed Model Predictive Control](#), *IFAC Workshop on Distributed Estimation and Control in Networked Systems (NecSys)*, 2019.
- [C3] X. Feng, **H. Hu**, M. E. Villanueva, and B. Houska. [Min-Max Differential Inequalities for Polytopic Tube MPC](#), *American Control Conference (ACC)*, 2019.
- [C2] **H. Hu**, Y. Pu, M. Chen, and C. J. Tomlin. [Plug and Play Distributed Model Predictive Control for Heavy Duty Vehicle Platooning and Interaction with Passenger Vehicles](#), *IEEE Conference on Decision and Control (CDC)*, 2018.

[C1] **H. Hu**, X. Feng, R. Quirynen, M. E. Villanueva, and B. Houska. [Real-Time Tube MPC Applied to a 10-State Quadrotor Model](#), *American Control Conference (ACC)*, 2018.

PREPRINTS

- [P3] R. Wu*, R. Yang*, D. D. Oh, D. P. Nguyen[†], and **H. Hu**[†]. [Turning Safety into Competence: Minimally Exploitable Robot Policies via Safety-Filtered Reinforcement Learning](#), 2026.
- [P2] D. D. Oh, D. P. Nguyen, G. Yuan, Q. Li, J. F. Fisac[†], and **H. Hu**[†]. [Winning a Won Game: Strict Reach-Avoid-Stay Control Barrier Functions for High-Dimensional Black-Box Systems](#), 2026.
- [P1] K. Liang, **H. Hu**, X. Zhao, D. Song, T. L. Griffiths, and J. F. Fisac. [Machine Bullshit: Characterizing the Emergent Disregard for Truth in Large Language Models](#), 2025.

TEACHING

Human-Centered Robotics: Models and Algorithms Johns Hopkins, Fall 2026
Instructor; EN.601.497/697 (Undergraduate and Graduate)

F1/10 Autonomous Racing Cars (Graduate) University of Pennsylvania, Spring 2026
Delivered a guest lecture on game theory and HRI; Instructor: Rahul Mangharam

Science and Art Summer School (Undergraduate) University of Padova, Summer 2024
Delivered a guest lecture on robotics and HRI; Course Director: Ernesto Caraffoli

Safety-Critical Robotic Systems (Graduate) Princeton University, Fall 2021, 2022
Head Teaching Assistant; Delivered guest lectures on robust motion planning; Instructor: Jaime F. Fisac

Model Predictive Control (Graduate) University of Pennsylvania, Spring 2020
Head Teaching Assistant; Delivered a guest lecture on invariant sets; Instructor: Manfred Morari

INVITED TALKS

University of Delaware, CIS Seminar 2026
 Game-Theoretic Safety and Agility for Human-Centered Robotics

UPenn GRASP Student Faculty Industry (SFI) Seminar 2025
 Johns Hopkins University, CS, ECE, & LCSR Joint Seminar 2025
 Duke University, Thomas Lord Department of MEMS Seminar 2025
 Texas A&M University, CS & ECE Joint Seminar 2025
 George Washington University, MAE Department Seminar 2025
 University of Colorado Boulder, ECEE Department Seminar 2025
 North Carolina State University, ECE Department Seminar 2025
 UCSD Safe Autonomous Systems Lab 2024
 From Gambits to Assurances: Game-Theoretic Integration of Safety and Learning for Human-Centered Robotics

IEEE CDC Workshop on Bifurcations for Control 2024
 Constructing Nonlinear Opinion Dynamics for Split-Second Game-Theoretic Interactions

Scania CV AB | KTH Royal Institute of Technology 2024
 The Safety Filter: A Unified View of Safety-Critical Control in Autonomous Systems

University of Padova Science and Art Summer School 2024
 Robotics: Toward Safer and More Efficient Interaction with Humans

Northeast Systems and Control Workshop 2024
 Human-Aware Safety Filters: A Unified View of Safety-Critical Control, Learning, and Interaction

UPenn Vijay Kumar Lab CMU Safe AI Lab UIUC Human-Centered Autonomy Lab	2024
Who Plays First? Optimizing the Order of Play in Stackelberg Games with Many Robots	
Creative Convergence Workshop	2023
Nonlinear Opinion Dynamics Meets Game Theory: Split-second Decisions for Interactive Robotics	
Toyota Research Institute	2023
Learning to Break Indecisions in Multi-Agent Interaction	
University of Colorado Boulder Honda Research Institute USA, Inc.	2022
Active Uncertainty Reduction for Human–Robot Interaction	
Princeton University, Robotics Seminar	2021
Shielding-Aware Robust Planning for Safe and Efficient Human–Robot Interaction	
ETH Zürich, Melanie Zeilinger Lab	2020
Distributed Model Predictive Control for the Internet-of-Things	
University of Pennsylvania, George Pappas Lab	2018
Plug and Play Distributed MPC for Heavy Duty Vehicle Platooning	
Chinese Academy of Sciences, Shanghai Branch	2018
Real-time Tube-based Model Predictive Control	
UC Berkeley, Hybrid Systems Laboratory	2017
Model Predictive Control for Fast and Guaranteed Safe Motion Planning	

ADVISING AND MENTORSHIP

Ph.D. advising, Johns Hopkins University (2026 - present)

Yuanhong Zeng (PhD student at JHU CS)

Zizhe Zhang (PhD student at JHU CS)

Geeta Chandra Raju Bethala (PhD student at JHU CS, co-advised with Krishna Murthy)

Shereiff Garrett (PhD student at JHU CS, co-advised with Krishna Murthy and Mathias Unberath)

Doctoral research mentoring

Wei Wang (PhD student at Penn CIS)

Donggeon David Oh (PhD student at Princeton ECE)

Justin Wang (PhD student at Princeton ECE)

Madison Bland (PhD student at Princeton ECE)

Master's research mentoring

Honghao Zhao (JHU CS)

Siwen Hu (JHU CS)

Qingchen Li (JHU CS)

Botong Qu (JHU Robotics)

Chengyu Xiao (JHU CS)

Albert Gimó Contreras (Visiting Student at Princeton, recipient of the “La Caixa” Fellowship)

Xinyi Liu (UMich/Honda Research Institute → PhD at UNC/NCSU Joint Department of BME)

Undergraduate students

Nathan Baek (JHU CS)

Ruihan Wu (Visiting Student at JHU CS)

Rui Yang (Visiting Student at JHU CS)

Gongkai Yuan (Visiting Student at JHU CS)

Yicong Lei (Visiting Student at JHU CS)

Yanbin Wang (Visiting Student at JHU CS)
 Fanxiu Sophie Qiu (JHU CE & Neuroscience)
 Kensuke Nakamura (Princeton ECE → PhD student at CMU Robotics Institute)
 Vikash Modi (Princeton ECE → SpaceX)

OUTREACH **Program Co-organizer, [Princeton AI4ALL](#)** 2022 - 2024
The summer camp teaches AI technology and policy to high school students, to broaden access to AI and inspire our future leaders to use it ethically.

Lecturer in Robotics, [University of Padova Science and Art Summer School](#) 2024
The summer school offers broad topics in science and art to undergraduate students in China and Italy.

OPEN-SOURCE SOFTWARE **[Safety-SB3](#)**: Scalable safety and reach-avoid reinforcement learning with Stable-Baselines3.

[Provably-Optimal-RL](#): Training and evaluation code for RL under safety filtering.

[RLHS](#): Preference generation and language-model fine-tuning with hindsight simulation.

[BranchAndPlay](#): An efficient solver that finds the socially-optimal Stackelberg (leader-follower) equilibrium for general-sum dynamic games.

[KLGame](#): An efficient solver for non-cooperative dynamic games with Kullback-Leibler (KL) regularization with respect to a general, stochastic, and possibly multi-modal reference policy.

[DeceptionGame](#): Approximate HJI Reachability analysis for robot safety in joint physical-belief spaces using adversarial reinforcement learning.

[OpinionGame](#): Automatic synthesis of nonlinear opinion dynamics using dynamic games.

[JaxRacing](#): JAX-based implementation of ILQR for safe autonomous racing.

[DualControlHRI](#): Active uncertainty reduction for human-robot interaction using dual control.

[SHARP](#): A human-aware motion planner that preempts costly safety-enforcing maneuvers triggered by unlikely human actions.

[LearningCBFs](#): Learning a provably valid control barrier function (CBF) from expert data.

MEDIA COVERAGE **[Safely scale AI agents to the physical world](#)** The Daily Pennsylvanian, 2026

[Game-theoretic integration of safety, interaction, and learning](#) AIhub, 2025

[Machine Bullshit: Disregard for truth in LLMs](#) New Scientist & IEEE Spectrum, 2025

[Data drives quicker, safer decisions for race cars and robots](#) Princeton EQuad News, 2024

[Who plays first? Using game theory to coordinate many robots](#) Princeton Research, 2024

SERVICE & PROFESSIONAL ACTIVITIES **Associate Editor**

 • IEEE International Conference on Robotics and Automation (ICRA) 2027

 • IEEE Robotics and Automation Letters (RA-L) 2024 - 2027

 • IEEE Conference on Advanced Robotics and its Social Impact (ARSO) 2026

Workshop & Seminar Organization

 • [Workshop on Safety Analysis for Autonomous Systems: Practical Perspectives and Recent Advances \(SAAS\)](#), Organizer 2026

 • [3rd Workshop on Multi-Agent Dynamic Games \(MAD-Games\)](#), Lead organizer 2026

 • [ACM/IEEE HRI Pioneers Workshop](#), Program Chair 2025

 • [ICRA Workshop on Public Trust in Autonomous Systems](#), Lead organizer 2025

- [Princeton Robotics Seminar](#), Organizer

2024 - 2025

Panelist & Reviewer for Grant Proposals

- NSF CMMI, Panelist 2026
- NSF CISE, Panelist 2026
- Schmidt Sciences, Reviewer 2026

Reviewer for Journals

- International Journal of Robotics Research (IJRR)
- IEEE Transactions on Robotics (T-RO)
- Automatica
- IEEE Transactions on Automatic Control (TAC)
- IEEE Transactions on Control Systems Technology (TCST)
- IEEE Transactions on Control of Network Systems (TCNS)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Control Systems Letters (L-CSS)
- International Journal of Robust and Nonlinear Control
- Optimal Control Applications and Methods
- IET Control Theory & Applications
- Chemical Engineering Science
- IEEE Access

Reviewer for Conferences and Workshops

- Robotics: Science and Systems (RSS)
- RSS Pioneers, Faculty Meta Reviewer
- Conference on Robot Learning (CoRL)
- World Symposium on the Algorithmic Foundations of Robotics (WAFR)
- ACM/IEEE International Conference on Human–Robot Interaction (HRI)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Conference on Decision and Control (CDC)
- American Control Conference (ACC)
- IFAC World Congress
- Annual Learning for Dynamics & Control Conference (L4DC)
- Conference of the International Association for Safe & Ethical AI (IASEAI)
- International Conference on Neuro-symbolic Systems (NeuS)
- IFAC Conference on Networked Systems (NecSys)
- Annual Conference on Information Sciences and Systems (CISS)
- RSS Safe Autonomy Workshop
- ICRA Workshop on Bridging the Lab-to-Real Gap